Experiment No. 1

Code:

```
import numpy as np
c1 = np.array([1, 1, 1, 1])
c2 = np.array([1, -1, 1, -1])
c3 = np.array([1, 1, -1, -1])
c4 = np.array([1, -1, -1, 1])
print("Enter the data bits (use 1 or -1):")
d1 = int(input("Enter D1: "))
d2 = int(input("Enter D2: "))
d3 = int(input("Enter D3: "))
d4 = int(input("Enter D4: "))
r1 = c1 * d1
r2 = c2 * d2
r3 = c3 * d3
r4 = c4 * d4
resultant_channel = r1 + r2 + r3 + r4
print("Resultant Channel:", resultant_channel)
Channel = int(input("Enter the station to listen (C1=1, C2=2, C3=3, C4=4): "))
if Channel == 1:
    rc = c1
elif Channel == 2:
    rc = c2
elif Channel == 3:
    rc = c3
elif Channel == 4:
  rc = c4
else:
    print("Invalid channel selected!")
    exit()
res = np.dot(resultant_channel, rc)
data = res / len(rc)
```

```
print("Recovered Data Bit:", int(data))
```

Output:

```
Enter the data bits (use 1 or -1):
Enter D1: 1
Enter D2: -1
Enter D3: 1
Enter D4: -1
Resultant Channel: [0 4 0 0]
Enter the station to listen (C1=1, C2=2, C3=3, C4=4): 2
Recovered Data Bit: -1
```